This summer, I worked primarily in Professor Tim Kovachy's lab to help design and test a new Strontium (Sr) vapor cell for laser locking in the lab's Magnetic Optical Trap (MOT). Due to Sr's low vapor pressure, one of the most significant difficulties I worked on was finding a way to get Sr to high enough temperatures to enter the gaseous phase as well as keep the chamber windows high enough temperatures to prevent build-up which would make laser locking using Saturated Absorption Spectroscopy (SAS) more difficult. This involved me first taking a pre-existing window design and creating prototypes to test how wrapping Nichrome (Nickel-Chromium alloy) wire around it and passing a current through it could heat the window. We chose Nichrome because of its high resistance (1.055 ohms/feet) allowing it to reach temperatures of around 1000 ℃ for the 22 AWG wire we chose.

Based on early tests, we discovered we needed to slightly redesign the window to hold the wire tightly around it, maximize the number of coils we can get, and find an electrically insulating insulation that is as thermally conductive as possible. We found that the primarily silicate mineral, Mica, would be the best insulator as it is both electrically insulating and more thermally conductive than other electrically insulating materials. With this new insulation, I reached temperatures of ~330 ℃ in the air using only 6 coils and 5 amps of current, which should be high enough to prevent the Sr from sticking and building up.

Next, I worked on designing a Sr reservoir that would hold the Sr and heat it enough to get into the gaseous phase and fill the vapor cell. This required a much higher temperature of around 500 ℃ and required me also to consider the thermal loads. So, in my design, I needed to include a stack design alternating between ceramic pieces and aluminum disks to reduce conductive heat loads (transfer of heat from direct contact) and include a heat shield to minimize radiative heat loads (transfer of heat with no contact due to thermal radiation).

After having a design, I made a prototype of the Sr reservoir, which I ran a similar heating test on using Mica and Nichrome like with the copper window. This time, I reached a max temperature of ~590 ℃ in the air using 13 coils and 5 amps. Additionally, during these tests, I varied the number of coils used to characterize how the number of coils affected max temperature and found a linear relationship.

Next, I ran a vacuum test on Mica to see if there was any outgassing (release of gasses trapped in the material once put in vacuum) we needed to worry about. First, I pumped down an empty vacuum and got a baseline reading using a Residual Gas Analyzer (RGA) of the partial

pressures. Then I added Mica and pumped it down again; taking a baseline revealed no outgassing at room temperature. However, since we would be heating it to incredible temperatures, I needed to rerun the test while heating the Mica. To do this, I set up the Sr Reservoir in the vacuum with 5 coils of Nichrome and Mica insulation. This also served as a way to compare the heating in air vs. vacuum. While heating, I reached a max temperature of 365 °C at 5 amps (in air 5 coils at 5 amps only reached 325 °C). The RGA showed that while there initially was a peak in gasses, it decreased and reached similar levels to our baseline over time. This implied that the mica tape we used initially had some material (likely the polymer backing) that outgassed but only when initially heated.

After running all these tests, I had to modify both designs and think about properly holding down the Nichrome to maximize contact and how to feed wire through to the Sr reservoir and window. And then, I had to get quotes for all the parts I would need for the heating parts of the reference cell.

While I didn't get as far as I hoped to start conducting saturated absorption spectroscopy this summer, I got lots of experience and practice in many areas that will be useful in future research. I gained a lot of experience with general heating, such as thinking about heat loads, heating elements, and insulations. I gained experience with reference cells and atomic ovens. I learned how to effectively use Solid Works to design and draw parts and then how to prototype those parts. I learned how to set up vacuum systems and how to use an RGA to analyze the gasses present in a vacuum.

I am now starting my transition to Professor Odom's lab, where my knowledge of heating and design will be helpful when designing/implementing a Rb oven and doing heat load calculations and simulations on various parts. I will also continue working in Professor Kovachy's lab to finish the project I started over the summer by testing the next version of the Sr reservoir, putting together the reference cell, and setting up SAS.